Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part d.ii in this mark scheme extract

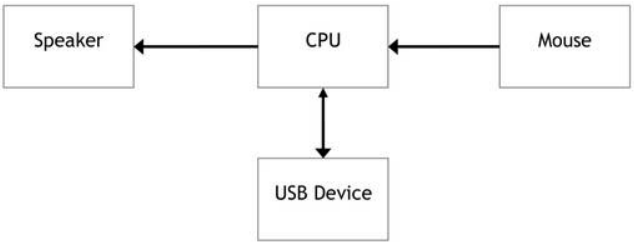
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part d.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part d.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · Write a logic statement for a problem

Answer 1 of 4 · Question 5 part d.ii · 3 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Write a logic statement for a problem

Answer 2 of 4 · Question 5 part b · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

5(a)	<table><tr><th>P</th><th>Q</th><th>R</th><th>Q OR R</th><th>P AND (Q OR R)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • All possible combinations in columns P, Q and R (1) • Correct values in Q or R (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 2.	3

Question Number	Answer	Additional Guidance	Mark
5(b)	B AND S AND NOT L Award **one** mark for each of: NOT L or B AND S (1) Fully correct (1)	• Any order and ignore brackets	2

Question Number	Answer	Additional Guidance	Mark
5(c)	Any **two** of: • Remove code vulnerabilities in programming languages (1). • Eliminate bad programming practices (1).		2

Question Number	Answer	Additional Guidance	Mark
5(d)	Award **one** mark each for any of: • Copyright (1) • Patent (1) • Licensing (1) • Trademark (1)		3

Type MS · Write a logic statement for a problem

Answer 2 of 4 · Question 5 part b · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

16

Type MS · Write a logic statement for a problem

Answer 2 of 4 · Question 5 part b · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Write a logic statement for a problem

Answer 3 of 4 · Question 6 part c · 4 marks

2021 NOV · Paper 01 · extract 1 of 6 · find part c in this mark scheme extract

6(a)(i)	Award **one** mark from: • faster translation/execution (1) • allows her to directly address components / make efficient use of available memory space (1) • can dispense with the need for an OS to make the code run faster/free up memory space (1)		1
6(a)(ii)	Award **one** mark from: • difficult to read/understand (1) • easy to make mistakes (1) • can be hard to find errors in the code (1) • time consuming (to write) (1) • not very portable / processor-specific (1) • lack of built-in functions/procedures (1) • few development/editorial tools available (1)		1

Type MS · Write a logic statement for a problem

Answer 3 of 4 · Question 6 part c · 4 marks

2021 NOV · Paper 01 · extract 2 of 6 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark from: • virus detection/scanner (1) • threat identification (1) • real-time scanning (1) • scan scheduling (1) • quarantine files (1) • carry out automatic updates (1) • on-demand file scanning (1) • kill switch (1)		2
6(b)(ii)	Award **one** mark from: • anti-spyware (1) • anti-adware (1)		1
6(c)	A AND O AND (W OR D) Award **one** mark each to a maximum of **four** marks for: • A AND O (1) • W OR D (1) • Brackets around W OR D (1) • AND between A AND O (W OR D) (1)		4

Type MS · Write a logic statement for a problem

Answer 3 of 4 · Question 6 part c · 4 marks

2021 NOV · Paper 01 · extract 3 of 6 · find part c in this mark scheme extract

6(d)	**Intellectual property** • Intellectual property is any work that is distinct, owned, and protected by patent or copyright laws • Akiko's software programs fall into this category • She will be able to protect her intellectual property • Her software is protected by copyright without her having to apply for it **Licensing** • Purchasing software does not mean you own it • Could add a licence key to the software. • Could require compulsory registration using the internet • Could allow proprietary licences, which do not allow code modification or code reuse. • Could allow Free and open-source software (FOSS), which would allow the user to modify and reuse the code. • Creative Commons (CC) ○ Could use creative commons (CC) licences that would allow the free distribution of copyrighted work ○ Would use if Akiko wanted to allow people to share, use and build on the programs. ○ May mention some of the CC licences and conditions	6

Type MS · Write a logic statement for a problem

Answer 3 of 4 · Question 6 part c · 4 marks

2021 NOV · Paper 01 · extract 4 of 6 · find part c in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.
		Total **15**

Type MS · Write a logic statement for a problem

Answer 3 of 4 · Question 6 part c · 4 marks

2021 NOV · Paper 01 · extract 5 of 6 · find part c in this mark scheme extract

Type MS · Write a logic statement for a problem

Answer 3 of 4 · Question 6 part c · 4 marks

2021 NOV · Paper 01 · extract 6 of 6 · find part c in this mark scheme extract

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Type MS · Write a logic statement for a problem

Answer 4 of 4 · Question 5 part e · 3 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

5(a)	Award **one** mark for each correct column: **Characteristic** / **Translator**: Assembler, Interpreter, Compiler Does **not** have to be present in memory to execute the translated code. — ✓ Compiler Translates high level language source code. — ✓ Interpreter, ✓ Compiler Executes high level code until either the program has finished, or a syntax error is encountered. — ✓ Interpreter Translates low level language source code into machine code. — ✓ Assembler		3
5(b)	Award **two** marks for a linked explanation such as: • Fewer instructions are executed per second (1) so gameplay will not appear to execute as smoothly / graphics lag / fewer frames per second / can only handle less complex graphics (1) • Consoles will execute more instructions per second / have faster processors / GPUs (1) so gameplay will not be realistic / mirrored / fair comparison (1)	One mark can be awarded processing speed implication, but the second mark is dependent on having a gameplay impact being identified. e.g. "A lower clock speed means processing will take longer" is generic for one mark. Low performance on its own is not enough.	2

16

Type MS · Write a logic statement for a problem

Answer 4 of 4 · Question 5 part e · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

<table>
<tr>
<td>5(c)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Issue</th><th>Utility software</th></tr>
<tr><td>Her word processor cannot read all of the files because it is corrupt.</td><td>File repair/recovery</td></tr>
<tr><td>She has a JPEG image and needs to send it to a publisher who can only read PNG image files.</td><td>Converting</td></tr>
<tr><td>There has been a flood and files on a server have been damaged, but Anna needs to access her data.</td><td>Backup/restore</td></tr>
<tr><td>Anna's hard drive is running very slowly when she tries to access a file.</td><td>Defragmentation/defrag</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>5(d)</td>
<td>Award one mark for:
• Lossless (1)

Award one mark for any of the following up to a maximum of two marks:
• No data will be lost / the process is reversable (1)
• The manual will still be readable (1)
• Removal of data would change the meaning of text (1)
• Proofreader will be able to edit the text (1)</td>
<td>Allow converse if candidate gives reasons why lossy would not be used for justification

Allow RLE as a valid lossless compression method.</td>
<td>3</td>
</tr>
</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Write a logic statement for a problem

Answer 4 of 4 · Question 5 part e · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

5(e)	Award **one** mark for each of: Method 1 • (C AND NOT D) (1) • (NOT C AND D) (1) • <expr1> OR <expr2> (1) Method 2 • (C OR D) (1) • NOT (C AND D) (1) • <expr1> AND <expr2> (1) Example statements: (C AND NOT D) OR (NOT C AND D) (C OR D) AND NOT (C AND D)	Order of precedence is NOT, AND, OR Note other equivalent solutions MAX 2 if all terms correct apart from missing brackets that are required. Do not credit circuit diagrams.	3
		Total for question 5	**15**

Type MS · Write a logic statement for a problem

Answer 4 of 4 · Question 5 part e · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark